

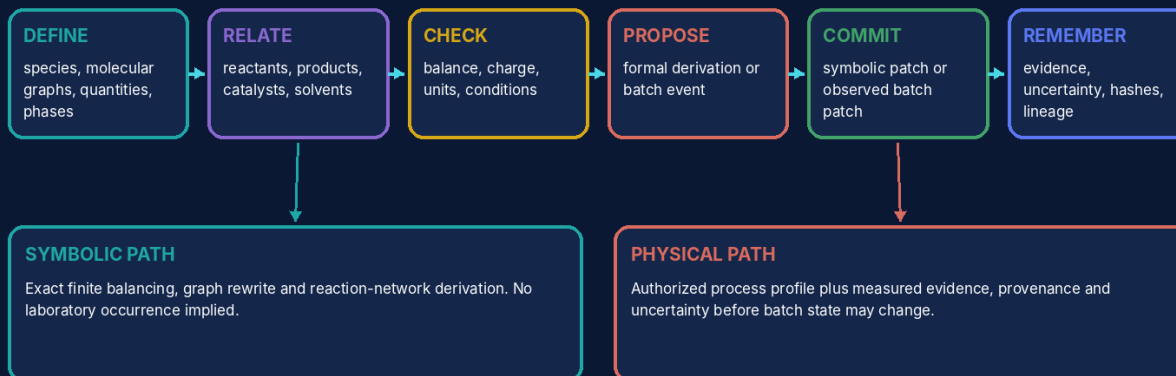
UNIVERSAL GEOMETRIC TOPOLOGICAL
OPEN MODULAR SUBSTRATE

UGTOMS REACT 1.0

Typed Reaction, Conservation, Conditions and Evidence Cartridge

UGTOMS REACT 1.0 architecture

Typed species, quantities and reaction rules remain upstream of physical batch authority.



Symbolic and physical authority are deliberately separate.

Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

ugtoms.domain.react@1.0.0

Component	REACT
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Status	Formal domain cartridge; specification self-checked; laboratory and regulatory validation not claimed

ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

Contents and Reading Rule

Section	Purpose
1	Executive decision and release scope
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READING RULE	
The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.	

Executive Decision and Release Scope

FORMAL DECISION

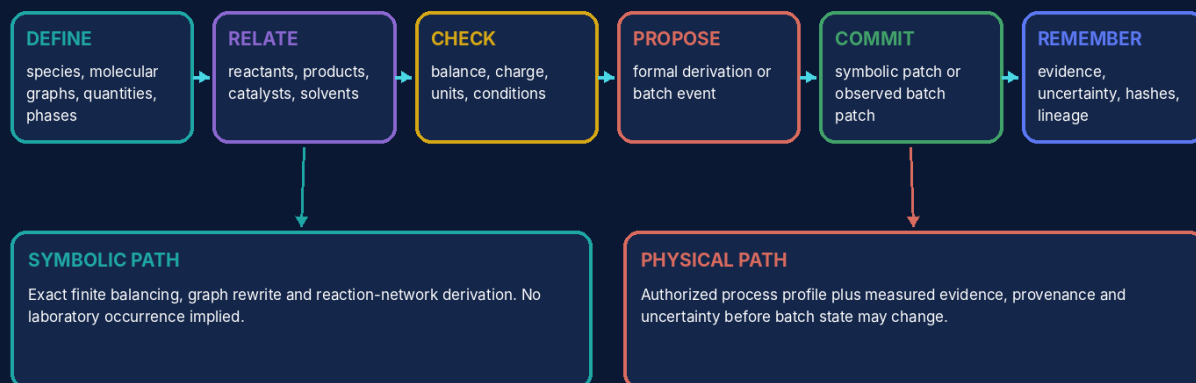
GO for REACT as the first UGTOMS specialist cartridge. Chemistry becomes a typed relation-and-event domain without pretending that a balanced equation, molecular drawing or simulation is an observed physical result.

REACT packages species identity, molecular graphs, quantities, units, phases, reaction relations, conditions, conservation invariants, hazards, evidence and lineage into the same open modular discipline used elsewhere in UGTOMS. Its most important design decision is a dual authority boundary:

- Formal derivation authority may balance equations, evaluate finite graph rewrites and traverse a reaction network.
- Physical batch authority may change only through an authorized, evidence-bearing event with measured quantities, conditions, uncertainty and provenance.
- Notation and glyphs resolve typed OperatorCells, but their visual form cannot manufacture reversibility, hazard or occurrence.

UGTOMS REACT 1.0 architecture

Typed species, quantities and reaction rules remain upstream of physical batch authority.



Symbolic and physical authority are deliberately separate.

Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

Figure 1. REACT maps chemistry-specific definitions into the retained UGTOMS authority chain.

Source Basis and Evidence Discipline

The cartridge inherits the following UGTOMS commitments:

- Typed definitions and operators live inside the substrate, with stable IDs, dependencies and content hashes [S1-S3].
- Pure evaluation is upstream of support, compatibility, guard, verified proposal, deterministic commit and lineage [S1, S2, S4].
- Coordinates, pictures, sounds, meshes and packed codes are not identity or authority by themselves [S4-S8].
- Uncertainty, exactness, capability and error margins are explicit and may only be strengthened by evidence [S1, S2, S8].
- Component-scoped identities prevent a specialist domain pack from masquerading as a universal replacement of every other release [S10].

ENGINEERING-DERIVED DOMAIN CONTENT

The chemistry type system, stoichiometric model, molecular-graph records, reaction notation semantics, hazard gates and application examples are original engineering extensions. No chemical database, laboratory protocol, safety data sheet, regulatory standard or synthesis corpus is imported by this PDF.

ID	Registered source	Use in this cartridge	Pages
S1	UGTS_FOUNDATION_UNICODE_v5.pdf	Foundation 5.0 OperatorCells, exactness/capability, Unicode literal/glyph/semantic separation, hot codebooks and authority boundary.	pp. 1-21
S2	UGTS_KC_4_2_GeneralOperator_Order_Addendum(1).pdf	Typed OperatorSpec, domains/codomains, units, laws, effects, dependencies, order profiles, certificates and open extension contracts.	pp. 2-18
S3	UGTS_KC_3_6_Tom_Klootwijk.pdf	Definitions as first-class content-addressed data, resolvable dependencies and explicit operation order.	pp. 1-14
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	Canonical support → compatibility → guard → event → transition → lineage sequence; packed-state and performance evidence discipline.	pp. 2-15
S5	UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	Pattern, field, topology, kinematics, bounded dynamics and hybrid-event architecture.	pp. 3-15
S6	UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	Stable identity, proposal verification, deterministic commit, checkpoints, replay and downstream presentation.	pp. 3-16
S7	UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	Versioned projects, deterministic game-world snapshots/hashes, compact records and self-contained delivery.	pp. 2-16
S8	UGTS_KC_4_0_Spatial_Evidence_Ledger(1).pdf	Observation versus authority, uncertainty, provenance, atomic patches, checkpoints and evidence-bearing lineage.	pp. 3-14
S9	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	Finite packing, log-polar charts, interval guards, winding and explicit topology/rotation distinctions.	pp. 1-15
S10	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	Component-scoped semantic identities and explicit release graphs instead of one misleading global number ladder.	pp. 1-6

Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_REACT:

D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)	
Field	Contract
T	Entity, value and state types
O	Typed operators and transformations
R	Relations, graphs, topology and converse pairs
U	Units, dimensions, coordinate and profile conventions
I	Invariants, conservation laws and constraints
Q	Dependency, precedence, phase and reduction order
E	Observations, evidence, uncertainty and provenance
G	Guards, thresholds and admission predicates
Delta	Proposed and committed state patches
L	Lineage, replay, history and migration
Cert	Exactness, residual, interval, scope and status certificates
K	Cold atlas, hot codebook and packed representation
Pi	Optional visual, audio, physical or device projection

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

General Operator Record and Order

```
OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape,
               laws, exactness, effects, capabilities, dependencies,
               order_contract, invariants, provenance, content_hash)
```

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

EXAMPLE RECORD

reaction.transform consumes a typed multiset of SpeciesAmount ports and returns a ProductProposal plus StoichiometryCertificate. Its surface arrow may be $\rightarrow$ or a registered Unicode arrow. Reverse evaluation is disabled unless the ReactionRule declares a supported reverse direction and condition profile.

Normative phase schedule

Phase	Meaning
1-3	parse, normalize and resolve content-addressed references
4-6	type, shape and unit validation; dependency closure
7-9	construct candidates; apply declared composition and reduction order
10-12	evaluate exact or bounded results; issue capability and residual certificate

Phase	Meaning
13	support, compatibility and guard classification
14	verified proposal and deterministic conflict resolution
15	atomic commit, lineage, replay and optional projection

Formal REACT State and Authority Split

q_REACT = (P, S, G, Q, C, R, B, E, L, U)

P profile/version S species registry G molecular graphs
 Q typed quantities C conditions R reaction rules
 B batch/mixture state E evidence/provenance L lineage/replay
 U uncertainty/status

Object	Authoritative role	Boundary
SpeciesDefinition	Stable identity, composition/graph, charge, phase capabilities and aliases.	A formula string or drawing is a representation, not identity by itself.
ReactionRule	Typed reactant/product ports, coefficients, atom map, conditions, invariants and capability.	A rule may be formal-only; it need not assert laboratory occurrence.
BatchState	Actual inventory/mixture quantities, conditions and accepted evidence.	Only evidence-bearing commits may mutate it.
ReactionProposal	Candidate derivation or batch event with requested direction and inputs.	Proposal is not change.
ReactionCertificate	Balance, charge, units, capability, residual and provenance scope.	Certificate scope is explicit and never upgraded silently.

DUAL AUTHORITY RULE

A symbolic product set may be exact relative to a ReactionRule while the physical product quantity remains unknown. REACT never promotes expected product, yield, rate, equilibrium position or safety into batch truth without the declared evidence path.

REACT dual-authority demonstration

A balanced symbolic derivation is not automatically a real batch observation.

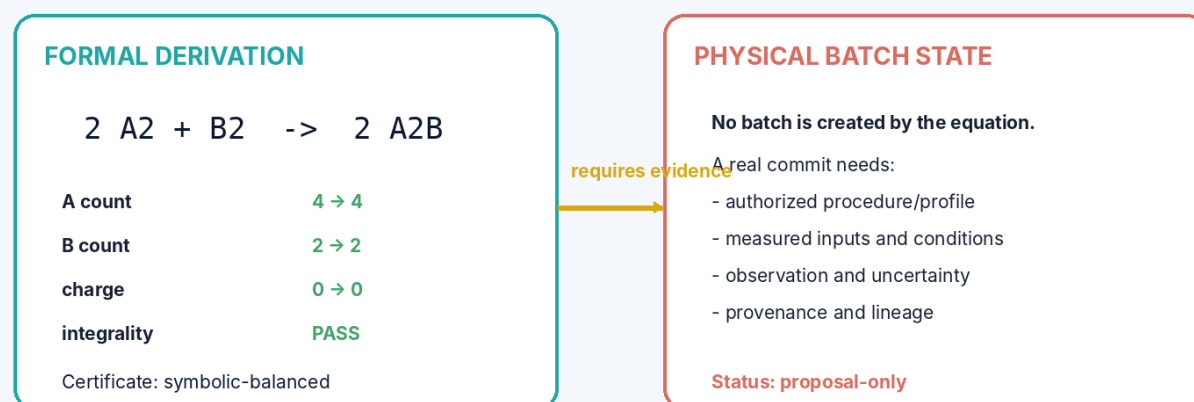


Figure 2. A balanced formal derivation remains proposal-only for physical batch state.

Species, Quantities and Reaction Rules

SpeciesDefinition

```
SpeciesDefinition = (id, version, aliases, formula_profile, element_vector,
                    isotope_vector?, formal_charge, molecular_graph?, phase_capabilities,
                    provenance, content_hash)
```

The finite element-count vector supports exact balancing when the formula profile is complete. Molecular graphs may be used for atom mapping and graph-rewrite templates, but graph equivalence, stereochemistry and physical identity require explicit capability profiles and external chemistry oracles.

Typed quantities and mixtures

Quantity	Required fields	Common failure rejected
Amount	value, unit, uncertainty, basis	mixing amount-of-substance with mass without a conversion adapter
Concentration	numerator, denominator, basis, unit, state context	using a bare number such as 5 without dimension or basis
Temperature	absolute unit and uncertainty	treating Celsius differences and absolute temperature as the same type
Pressure	unit, measurement context, uncertainty	assuming ambient pressure without declaring it
Mixture fraction	component, basis, normalization policy	combining mass, mole and volume fractions silently

ReactionRule

```
ReactionRule = (id, direction_profile, reactant_ports, product_ports, coefficients,
               atom_map?, charge_map, condition_window, catalysts, solvents,
               order_contract, invariants, capability, evidence_requirements, provenance, hash)
```

NOTATION BOUNDARY

The arrow expresses a registered reaction relation. It is not an animation path, causal proof or permission to execute a physical process. A mirrored arrow is not automatically the reverse rule.

Conservation, Conditions and Order

Exact finite stoichiometric check

For element matrix A and coefficient vector nu:
 $A * nu = 0$
 with reactant coefficients negative and product coefficients positive.
 Charge balance adds: $z \cdot nu = 0$.

The exact profile accepts integer coefficients and finite element/charge vectors. Fractional display coefficients may normalize to an integer vector. Null-space solving, canonical scaling and underdetermined systems are kept separate from experimental yield or mechanism claims.

Verified proposal predicate

```
verified(a,q) = in_support
               and type_compatible
               and unit_compatible
               and element_balance
               and charge_balance
               and coefficient_policy
               and condition_guard
               and hazard_policy
               and authorization
               and numeric_error <= event_margin
```

Result-affecting order

Order relation	Why it matters
normalize formula → resolve species → balance	balancing before alias/species resolution can combine unlike identities
mix proposal → condition update → reaction evaluation	evaluating before the condition profile may select the wrong rule direction
formal derivation → evidence adapter → physical commit	symbolic output cannot skip measurement and authorization
same-time proposals → deterministic conflict policy	two reactions cannot consume the same authoritative inventory twice

Capability ladder

Level	Meaning
C0 notation-only	Literal reaction syntax and typed ports; no transformation claim.
C1 exact finite stoichiometry	Element/charge/unit invariants over finite declared vectors.
C2 graph-rewrite template	Declared molecular graph pattern and atom map under bounded semantics.
C3 bounded model	Thermodynamic, kinetic or equilibrium result from a named model with residual/status.
C4 calibrated simulation	Model parameters tied to calibration/validation evidence.
C5 observed batch	Physical state supported by measurements, provenance and uncertainty.
C6 governed process	External regulatory, safety, authorization and qualified-process profile.

Unicode Reaction OperatorCells

Foundation 5.0 allows literal, semantic and geometric faces to share one address without collapsing them [S1]. REACT registers a chemistry notation atlas with ASCII fallbacks and explicit non-claims.

Literal / fallback	Typed meaning	Converse or inverse rule	Boundary
→ / reaction-right	direct reaction relation from normalized input ports to output ports	none unless rule declares a reverse direction	visual direction does not prove spontaneity or occurrence
← / reaction-left	surface spelling with reversed ports for a registered reverse relation	shares a kernel only when the pair is explicitly registered	not generated by mirroring an arbitrary rule
⇌ / equilibrium	reversible/equilibrium relation under a condition profile	surface ports may swap; thermodynamic quantities do not automatically invert	does not mean equal amounts
+ / co-reactant	typed multiset composition in a reaction side	commutative only for the selected multiset profile	not ordinary numeric addition
Delta / heat condition	condition marker referencing a heat/temperature profile	no generic inverse	not an operational instruction
h nu / photon condition	condition marker referencing a photonic profile	no generic inverse	frequency, intensity and exposure remain external data

KLEIN-CONVERSE BOUNDARY

REACT may reuse the Foundation 5.0 port-swap mechanism for a registered direct/reverse notation pair. The topology bit selects a surface and port map; it does not make an irreversible reaction reversible, conserve energy automatically or erase the two lineage events.

Hot codebook policy

- Cold identities are stable Unicode/ASCII OperatorCells plus content hashes.
- A local 8- or 16-slot hot codebook is valid only when bound to the cold-atlas hash.
- Unknown and reserved slots reject; they never alias a hazardous or existing operator.
- Packed reaction nodes carry only hot-path fields. Units, species definitions, uncertainty, hazard policy and lineage stay external.

Evidence Ledger, Hazards and Authorization

Record	Required fields	Authority rule
Observation	instrument/method, calibration, value, unit, uncertainty, time, batch, provenance	may support a proposal; cannot rewrite a batch directly
BatchEvent	proposal, accepted/rejected status, consumed/created quantities, pre/post hashes	sole authoritative inventory mutation path
ConditionEvent	temperature/pressure/phase/profile change plus evidence	may change which rules are supported
HazardDecision	policy ID/version, classification source, scope, reason code	external policy is required for operational use
Checkpoint	state hash and event sequence	replay must reproduce the checkpoint state

SAFETY-FIRST SCOPE

This specification does not provide synthesis recipes, scale-up parameters, energetic/toxic production instructions, controlled-substance workflows or hazard classifications. A production implementation must use authoritative external data and block operational commits when the policy adapter is absent.

Authorization predicate

```
physical_commit_allowed = authorized_profile
                        and hazard_policy_pass
                        and operator_capability >= required_capability
                        and evidence_scope_complete
                        and quantities_nonnegative
                        and conflict_free
```

Deterministic Demonstration

The reference fixture uses abstract species so it cannot be mistaken for a laboratory procedure:

```
proposal: 2 A2 + B2 -> 2 A2B
left totals: A=4, B=2, charge=0
right totals: A=4, B=2, charge=0
result: symbolic-balanced
physical batch mutation: NONE
```

Test	Expected result	Reason code / certificate
balance forward	ACCEPT symbolic derivation	element-balance; charge-balance; integer-coefficients
request reverse on a forward-only rule	REJECT	reverse-not-declared
omit B2 inventory in a physical proposal	REJECT	insufficient-authoritative-quantity
use unitless temperature	REJECT	unit-profile-missing
present balance certificate as hazard approval	REJECT	certificate-scope-mismatch
append a measured batch observation with provenance	ACCEPT evidence record	observation-accepted; no automatic reaction event

Queries

- `balance(reaction_rule) → StoichiometryCertificate`
- `supported_rules(batch_state, condition_profile) → ordered rule IDs plus rejection reasons`
- `reaction_neighbors(species_id, capability_floor) → formal network edges`
- `trace_batch(batch_id) → complete accepted/rejected event lineage`
- `compare_batches(a,b) → quantity/condition/evidence deltas with uncertainty`

Packing and external adapters

Layer	Recommended treatment
Hot reaction node	local species/operator slots, coefficient deltas, direction flag, condition-profile slot and integrity bits
Cold atlas	full SpeciesDefinition, ReactionRule, units, hazard policy, provenance and content hashes
Thermodynamics/kinetics	adapter to a named mature library/database; return model/version/residual/status
Hazards/regulation	adapter to authoritative jurisdiction/version-specific sources; never inferred from graph alone
Visualization	molecular drawing, network graph or animation remains downstream

Reference Validation, Boundaries and Kill Criteria

REFERENCE SELF-CHECK RESULT

24/24 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

Checked area	Result
Catalog integrity	48 unique RCT IDs; required family/name/contract fields present.
Authority split	Formal derivation fixture cannot mutate physical batch state.
Stoichiometry	Abstract A/B fixture balances exactly; unbalanced variant rejects.
Charge and units	Declared checks pass; missing/incorrect profiles reject.
Order and conflicts	Reverse-not-declared and double-consumption cases return deterministic reason codes.
Evidence scope	Hash/provenance and certificate-scope boundaries are explicit.

Explicit non-claims

- No laboratory reaction occurrence, yield, rate, equilibrium composition or mechanism is established.
- No chemical safety, environmental, medical, regulatory or industrial suitability is certified.
- No external chemical standards, databases or safety data sheets are bundled or silently paraphrased.
- No universal molecular equivalence, exact 3D geometry or complete chemistry ontology is claimed.
- No performance or compression advantage is claimed without an implementation and equal-error benchmark.

Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

Complete REACT 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

ID	Family	Mechanism	Core contract
RCT001	Identity	Species registry	Stable species IDs, aliases, formula profile and provenance remain distinct.
RCT002	Identity	Species alias map	Aliases resolve to a stable identity; spelling does not redefine composition.
RCT003	Identity	Element-count vector	Finite element/isotope multiplicity vector under a declared formula profile.
RCT004	Identity	Formal charge	Integer or bounded charge value with explicit convention.
RCT005	Identity	Phase profile	Solid, liquid, gas, aqueous or custom phase is typed and profile-bound.
RCT006	Identity	Molecular graph	Atoms/bonds form a finite labeled graph; graph identity is not a screen image.
RCT007	Quantity	Amount quantity	Typed amount with dimension, unit and uncertainty.
RCT008	Quantity	Mass quantity	Mass is not inferred from display geometry; molar-mass adapter is explicit.
RCT009	Quantity	Volume quantity	Volume uses declared temperature/pressure context where required.
RCT010	Quantity	Concentration quantity	Numerator, denominator, basis and unit are explicit.
RCT011	Quantity	Temperature condition	Absolute temperature type; no unitless substitution.
RCT012	Quantity	Pressure condition	Pressure type with profile and uncertainty.
RCT013	Relation	Reactant-of	Species-to-reaction input relation with stoichiometric coefficient.
RCT014	Relation	Product-of	Species-to-reaction output relation with stoichiometric coefficient.
RCT015	Relation	Catalyst-of	Condition/role relation; does not imply a universal kinetic law.
RCT016	Relation	Solvent-of	Mixture role relation with explicit basis.
RCT017	Relation	Phase-of	Species/batch phase relation at a declared state.
RCT018	Relation	Contained-in-batch	Batch membership relation with quantity and evidence links.
RCT019	Operator	Mix	Creates a mixture proposal; no reaction occurrence is implied.
RCT020	Operator	Separate	Creates a separation proposal under a declared process adapter.
RCT021	Operator	Reaction transform	Typed reactant graph/vector to product graph/vector relation.
RCT022	Operator	Reversible relation	Declares two supported directions; it is not inferred from a mirrored arrow.
RCT023	Operator	Heat condition marker	Attaches a temperature/energy condition profile; not an instruction by itself.
RCT024	Operator	Photon condition marker	Attaches a declared photonic condition profile.
RCT025	Invariant	Element balance	Element-count totals are equal across a declared reaction equation.
RCT026	Invariant	Charge balance	Declared formal-charge totals are equal.
RCT027	Invariant	Coefficient integrality	Selected exact profile requires finite nonnegative integer coefficients.
RCT028	Invariant	Unit dimension	Quantities may combine only under dimensionally valid operators.

ID	Family	Mechanism	Core contract
RCT029	Invariant	Batch nonnegativity	A commit cannot consume more authoritative inventory than is present.
RCT030	Invariant	Atom-map consistency	Optional atom mapping preserves declared atom identities across graph rewrite.
RCT031	Guard	Condition window	Temperature, pressure, phase, time and other declared conditions lie in range.
RCT032	Guard	Phase compatibility	Input phases satisfy the rule/profile.
RCT033	Guard	Catalyst presence	Required catalyst role is present and valid for the profile.
RCT034	Guard	Hazard policy	Hazard and operational restrictions pass an external policy gate.
RCT035	Guard	Authorization	Physical-process actions require an explicit authorized profile.
RCT036	Guard	Error margin	Numerical/model residual stays below the declared decision margin.
RCT037	Event	Formal derivation	Commits a symbolic reaction-network or balancing result only.
RCT038	Event	Batch observation	Commits measured batch facts with uncertainty and provenance.
RCT039	Event	Batch merge	Atomic inventory/mixture merge with pre/post hashes.
RCT040	Event	Batch split	Atomic split with quantity conservation and lineage.
RCT041	Event	Measurement	Appends a calibrated measurement record; it does not rewrite history.
RCT042	Event	Reason-coded rejection	Records the first failed gate without mutating authoritative state.
RCT043	Certificate	Stoichiometry certificate	Balance, charge, coefficient and formula-profile scope.
RCT044	Certificate	Capability certificate	Syntax, exact finite, bounded model, calibrated or observed capability.
RCT045	Certificate	Provenance certificate	Source, adapter, calibration, profile and content-hash lineage.
RCT046	Packing	Species hot codebook	Local code bound to a cold species/operator atlas hash.
RCT047	Query	Reaction-network query	Neighbors, reachable formal products, conserved quantities and evidence scope.
RCT048	Adapter	External chemistry oracle	Versioned thermodynamic, kinetic, hazard or standards library behind typed ports.

Claims Ledger

Claim	Disposition	Reason
A balanced equation proves the reaction will occur.	REJECT	Balance is a necessary symbolic invariant, not experimental occurrence, rate, yield or safety.
A reverse arrow is the inverse of every reaction.	REJECT	Reverse evaluation is admitted only when a reversible profile and conditions are declared.
Formula or molecular graph alone determines hazard.	REJECT	Hazard classification requires authoritative external data, context, concentration, route and governance.
Exact element vectors support exact stoichiometric balance.	ADMIT	Under a declared finite formula profile, element totals can be checked exactly.
Declared integer formal charges support exact charge balance.	ADMIT	The result is exact within the chosen charge convention.
A catalyst symbol supplies a universal rate law.	REJECT	Catalyst role and kinetic effect are model- and condition-dependent.
The equilibrium arrow means equal quantities.	REJECT	It denotes a reversible/equilibrium relation profile, not equal concentrations or rates by itself.
A seed can reconstruct an unstored experiment.	REJECT	Real measurements, uncontrolled inputs and uncertainty remain evidence records.
A content hash proves scientific truth or ownership.	REJECT	It commits to canonical bytes under a policy only.
Formal derivation and observed batch state are different authorities.	ADMIT	Symbolic state may propose expectations; physical state requires evidence-bearing commit.
Packed codes can replace units, uncertainty and lineage.	REJECT	Cold definitions and certificates remain external to hot codes.
This cartridge is a complete chemistry theory.	REJECT	It is a bounded domain ABI with adapters for mature science and standards.

High-Impact Application Profiles

Profile	What the cartridge enables
Balancing and reaction-network learning	A tactile or digital rule world where coefficients, species ports and conservation certificates are manipulable objects.
Evidence-bearing laboratory notebook	Observations, calibration, uncertainty, condition profiles and batch lineage share a deterministic ledger.
Process simulation shell	Thermodynamic or kinetic oracles can be attached without letting model estimates masquerade as observations.
Materials and formulation provenance	Track mixture versions, quantity basis, source evidence and accepted/rejected transformations.
Safety-governed automation interface	Operational events remain disabled until external hazard, authorization and qualified-process gates pass.

Final Formal Definition

FORMAL DEFINITION

UGTOMS REACT 1.0 is a component-scoped chemistry cartridge in which species, molecular graphs, quantities, phases, reaction relations, conditions, conservation invariants, evidence and hazards are typed substrate data; formal derivations and physical batch observations follow separate authority paths; and no symbolic, visual or packed representation may bypass support, compatibility, guards, deterministic commit or lineage.

Implementation sequence

- Freeze a JSON Schema and canonical hash policy for SpeciesDefinition, ReactionRule, BatchState and certificates.
- Implement exact finite stoichiometric/charge/unit validation as the scalar oracle.
- Add a safe abstract reaction-network demonstration and independent verifier.
- Attach read-only external chemistry/hazard adapters before any real-world process control.
- Promote to an evidence-bearing application only after laboratory, safety and jurisdiction-specific review.

RELEASE CONCLUSION

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.